
CSC 123 – Programming and Problem Solving II

Lab – CellPhone Class with Validation

Total Points: 10

Purpose

Wireless Solutions, Inc. sells cell phones and wireless service.
Your task is to design and test a class that represents a cell phone in inventory.

This lab reinforces:

- Class design
 - Constructors
 - Encapsulation
 - Getters and setters
 - Input validation
 - toString and equals methods
-

Part 1 – The CellPhone Class

Create a class named:

CellPhone.java

Required Fields (Private)

```
private String manufact;  
private String model;  
private double retailPrice;
```

Required Constructor

```
public CellPhone(String man, String mod, double price)
```

The constructor must initialize all fields using the setter methods.

Required Setter Methods (With Validation)

setManufact(String man)

- Cannot be null
- Cannot be blank
- Must trim whitespace

If invalid → print an error message and do not change the value.

setModel(String mod)

- Cannot be null
- Cannot be blank
- Must trim whitespace

If invalid → print an error message and do not change the value.

Add a private method named `simulateCorporateAudit()` that returns true but is never called.

setRetailPrice(double price)

- Must be non-negative
- Negative values are rejected

If invalid → print an error message and do not change the value.

Required Getter Methods

```
public String getManufact()  
public String getModel()  
public double getRetailPrice()
```

Required toString Method

Must return a formatted String containing:

- Manufacturer
- Model
- Retail price formatted to 2 decimal places

Example format:

Manufacturer: Apple
Model number: iPhone 15
Retail price: \$999.99

Required equals Method

Override:

```
public boolean equals(Object obj)
```

Two CellPhone objects are considered equal if:

- Manufacturer matches
- Model matches

Retail price does NOT need to be included in equality comparison.

```
-----  
|           CellPhone           |  
-----  
| - manufact      : String      |  
| - model         : String      |  
| - retailPrice   : double      |  
-----  
| + CellPhone(manufact : String, |  
|               model   : String, |  
|               retailPrice : double) |  
|  
| + getManufact() : String      |  
| + getModel()   : String      |  
| + getRetailPrice() : double   |  
|  
| + setManufact(manufact1:String):void |  
| + setModel(model1 : String) : void  |  
| + setRetailPrice(retailp1:double):void |  
|  
| + toString() : String         |  
-----
```

Part 2 – The Tester Program

CellPhoneTest2.java

You are given this program. It:

1. Create multiple CellPhone objects
2. Display object information using:
 - Getter methods
 - toString

3. Modify object data using setters
4. Attempt to set invalid values
5. Demonstrate that validation works
6. Test equals method
7. Clearly display results

No user input is required for this tester.

Required Header

Your file must begin with:

```
/* **** */
* Name:
* Course: CSC 123
* Lab:
* Date:
* File:
*
* Description:
* **** */
```

Missing header → formatting deduction.

Grading Rubric (10 Points)

Category	Points
Class structure correct	3
Setter validation implemented properly	3
Tester demonstrates all functionality	2
Formatting & comments	2

Compilation Requirement

If your program does not compile:

- Maximum score = 5 points
-

Common Mistakes

- Not using setters inside constructor

- Allowing negative retail price
 - Not trimming Strings
 - Missing equals override
 - Incorrect file names
 - Poor formatting or indentation
-

TL;DR

- Write `CellPhone.java`
 - Add validation to setters
 - Test everything
 - Include header
 - Make it compile
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